Jing Zhang

Curriculum Vitae
(updated on November 6, 2026)

New York University
Brooklyn, NY
E *z.jing@nyu.edu*
https://jingz6676.github.io/

Research Focus

I develop methods for **self-improving robots** that connect structural reasoning with skill learning. My long-term vision is **physical recursive self-improvement**: robots that build, assemble, and repair robots, expanding the capabilities that support the next cycle of learning and construction.

Professional Experience

New York University

Since 2026 **Faculty Fellow**, *Center for Robotics and Embodied Intelligence*.

2025–2026 **Adjunct**, *Department of Mechanical and Aerospace Engineering*.

2023–2026 **Postdoctoral Researcher**, *Department of Mechanical and Aerospace Engineering*.
Advisor: Chen Feng, AI4CE Lab.

2023–2024 **Postdoctoral Researcher**, *Department of Anthropology*.
Advisor: Radu Iovita, Anthrotopography Lab.

Education

Wuhan University

2023 **Ph.D.**, Photogrammetry and Remote Sensing.
Advisors: Bin Luo and Yajun Wang.
Thesis: "Deep-Learning-based Method for Fringe Projection 3D Imaging".

2019 **M.Sc.**, Photogrammetry and Remote Sensing.

2017 **B.E.**, Photogrammetry and Remote Sensing.

Grants & Awards

Honors

2025 **MIT Rising Stars in EECS.**

Awarded Research Grants

2026–2029 **Fragmented Legacy: Turning Broken 3D Material Evidence into Testable Archaeological Hypotheses** *Schmidt Sciences (HAVI Program)*, $800,000.00 over 36 months (plus $5,000 for PI travel). **Awarded, 2026**. Role: **Co-Principal Investigator**. *Contribution: Formulated the research concept and authored the full proposal.*

2025–2028 **OAC: DigitizedNYC: Large-Scale 4D Urban Digital Twin for Embodied AI** *National Science Foundation (NSF)*, $599,806.00. Role: **Co-Principal Investigator**. *Contribution: Formulated the research concept and authored the full proposal.*

2025–2026 **Enabling Functional Dexterous In-Hand Reorientation for General-Purpose Robotic Manipulation**, *NYU Postdoctoral Research and Professional Development Support Grants*, $2,498.00. Role: **Principal Investigator**. *Contribution: Formulated the research concept and authored the full proposal.*

2025–2026 **Towards Intelligent Human–Robot Collaborative Assembly with Long Video Understanding**, *NYU Research and Innovation Seed (RAISE) Grants*, $14,527.00. Role: **Co-Principal Investigator**. *Contribution: Formulated the research concept and authored the full proposal.*

Proposal Writing Contributions

2024–2025 **SCC-PG: Democratizing AI-based Urban Mobility Data Acquisition for Underserved Communities by Battery-powered Smart Cameras**, *National Science Foundation (NSF)*, $150,000.00.

Role: Main contributor to proposal writing and technical framing.

2024–2025 **Open Digital Twins for NYC**, *NVIDIA*, NVIDIA RTX 6000 Ada Graphics Cards (valued at $54,400).

Role: Main contributor to proposal writing and resource planning.

Participation in Funded Projects

2023–2024 **BCS: Understanding and Classifying Lithic Use Wear: A Systematic Study Using Controlled Tribological Experiments and Computer Vision**, *National Science Foundation (NSF)*, $213,247.00.

Role: Served as Postdoctoral Researcher, responsible for research execution and implementation.

Publications

Conference Papers (**bold** = Jing Zhang; * = corresponding author)

2026 Zeyu Jiang, Sihang Li, Siqi Tan, Chenyang Xu, Juexiao Zhang, Julia Galway-Witham, Xue Wang, Scott A. Williams, Radu Iovita, Chen Feng*, **Jing Zhang***, "**CRAG: Can 3D Generative Models Help 3D Assembly?**," *In Proceedings of the International Conference on Machine Learning (ICML)*, 2026. Official paper. Led the overall project and conceptualization of the core ideas.

2026 Xinhao Liu, Jiaqi Li, Youming Deng, Ruxin Chen, Yingjia Zhang, Yifei Ma, Li Guo, Yiming Li, **Jing Zhang***, Chen Feng*, "**Wanderland: Geometrically Grounded Simulation for Open-World Embodied AI**," *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. Official paper. Led the overall project and conceptualization of the core ideas. (**Highlight**)

2026 Heyang Yu, Yinan Han, Xiangyu Zhang, Baiqiao Yin, Bowen Chang, Xiangyu Han, Xinhao Liu, **Jing Zhang**, Marco Pavone, Chen Feng, Saining Xie, Yiming Li, "**Thinking in 360°: Humanoid Visual Search in the Wild**," *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. Official paper. Contributed to the writing and refinement of the manuscript.

2025 Sihang Li, Zeyu Jiang, Grace Chen, Chenyang Xu, Siqi Tan, Xue Wang, Irving Fang, Kristof Zyskowski, Shannon P. McPherron, Radu Iovita, Chen Feng*, **Jing Zhang***, "**GARF: Learning Generalizable 3D Reassembly for Real-World Fractures**," *In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. Official paper. Led the overall project and conceptualization of the core ideas.

2025 Irving Fang, Kairui Shi, Xujin He, Siqi Tan, Yifan Wang, Hanwen Zhao, Hung-Jui Huang, Wenzhen Yuan, Chen Feng*, and **Jing Zhang***, "**FusionSense: Bridging Common Sense, Vision, and Touch for Robust Sparse-View Reconstruction**," *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2025. Official paper. Led the overall project and conceptualization of the core ideas.

2025 Xinhao Liu, Jintong Li, Yicheng Jiang, Niranjan Sujay, Zhicheng Yang, Juexiao Zhang, John Abanes, **Jing Zhang**, Chen Feng, "**CityWalker: Learning Embodied Urban Navigation from Web-Scale Videos**," *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. Official paper. Led the paper writing and the formulation of its technical narrative.

2025 Sihang Li, Siqi Tan, Bowen Chang, **Jing Zhang**, Chen Feng, Yiming Li, "**Adversarial Exploitation of Data Diversity Improves Visual Localization**," *In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. Official paper. Contributed to the writing and refinement of the manuscript.

2024 **Jing Zhang**, Irving Fang, Hao Wu, Akshat Kaushik, Alice Rodriguez, Hanwen Zhao, Juexiao Zhang, Zhuo Zheng, Radu Iovita, and Chen Feng, "**LUWA Dataset: Learning Lithic Use-Wear Analysis on Microscopic Images**," *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 22563–22573. IEEE Computer Society, 2024. Official paper. (**Highlight**)

Journal Papers

2024 Fuqian Li, Xingman Niu, **Jing Zhang**, Qican Zhang, Yajun Wang, "**Physics-based Supervised Learning Method for High Dynamic Range 3D Measurement with High Fidelity**," *Optics Letters*, vol. 49, pp. 602–605, 2024. DOI: https://doi.org/10.1364/OL.506775. Contributed to the writing and refinement of the manuscript.

2023 **Jing Zhang**, Bin Luo, Fuqian Li, Xingman Niu, Qican Zhang, Yajun Wang, and Xiangcheng Chen, "**Single-Exposure Optical Measurement of Highly Reflective Surfaces via Deep Sinusoidal Prior for Complex Equipment Production**," *IEEE Transactions on Industrial Informatics*, vol. 19, no. 2, pp. 2039–2048, 2023 (first published online in 2022). DOI: 10.1109/TII.2022.3185660.

2022 Sijie Zhu, Zhoujie Wu, **Jing Zhang**, Qican Zhang, Yajun Wang, "**Superfast and Large-Depth-Range Sinusoidal Fringe Generation for Multi-Dimensional Information Sensing**," *Photonics Research*, vol. 10, pp. 2590–2598, 2022. DOI: https://doi.org/10.1364/PRJ.468658. Contributed to the writing and refinement of the manuscript.

2021 **Jing Zhang**, Bin Luo, Zhuolong Xiang, Qican Zhang, Yajun Wang, Xin Su, Jun Liu, Lu Li, and Wei Wang, "**Deep-Learning-based Adaptive Camera Calibration for Various Defocusing Degrees**," *Optics Letters*, vol. 46, pp. 5537–5540, 2021 (**Highlighted as an Editor's Pick**). DOI: https://doi.org/10.1364/OL.443337.

2021 **Jing Zhang**, Bin Luo, Xin Su, Lu Li, Beiwen Li, Song Zhang, and Yajun Wang, "**A Convenient 3D Reconstruction Model based on Parallel-Axis Structured Light System**," *Optics and Lasers in Engineering*, vol. 138, Art. no. 106366, 2021. DOI: https://doi.org/10.1016/j.optlaseng.2020.106366.

2020 Chengyuan Li, Bin Luo, Hailong Hong, Xin Su, Yajun Wang, Jun Liu, Chenjie Wang, **Jing Zhang**, Linhai Wei, "**Object Detection based on Global-Local Saliency Constraint in Aerial Images**," *Remote Sensing*, vol. 12, Art. no. 1435, 2020. DOI: https://doi.org/10.3390/rs12091435. Contributed to the writing and refinement of the manuscript.

2019 **Jing Zhang**, Bin Luo, Xin Su, Yuwei Wang, Xiangcheng Chen, and Yajun Wang, "**Depth Range Enhancement of Binary Defocusing Technique based on Multi-Frequency Phase Merging**," *Optics Express*, vol. 27, pp. 36717–36730, 2019. DOI: https://doi.org/10.1364/OE.27.036717.

2018 Yajun Wang, **Jing Zhang**, and Bin Luo, "**High Dynamic Range 3D Measurement based on Spectral Modulation and Hyperspectral Imaging**," *Optics Express*, vol. 26, pp. 34442–34450, 2018. DOI: https://doi.org/10.1364/OE.26.034442. Contributed to the code, writing, and refinement of the manuscript.

Teaching

New York University

Fall 2026 **Mathematics for Robotics** (ROB-GY 6013), *Graduate Course*, Instructor.

Fall 2025 **Robot Perception** (ROB-GY 6203), *Graduate Course*, Instructor.

Spring 2025 **Robot Vision** (ROB-UY 3203), *Undergraduate Course*, Co-Instructor.

University Service

New York University

2024 **Advisor for C.A.A.B. - CMML, Anthrotopography, AI4CE, and Bromage Labs**, **NYU Applied Research In Science and Engineering (ARISE) Program for High School Students.**

Project: Robot vs. Human Wear for Lithic Use-Wear Analysis.

2023 **Advisor for NYU Self Drive Team**, *NYU Multidisciplinary Undergraduate Vertically Integrated Project (VIP)*.
Project: Deep Gaussian Splatting Mapping.

Professional Service

Workshop Organizer

2026 CVPR, 2nd International Workshop on Spatial Intelligence for Cultural Heritage

2026 CVPR, Computer Vision for Science 2026

Conference Reviewer

NeurIPS, ICML, ICLR, CVPR, ECCV, ICCV, ICRA.

Journal Reviewer

Since 2024 Scientific Reports, IEEE Robotics and Automation Letters (RA-L), PLOS

Since 2021 Optics Letters, Optics and Lasers in Engineering

Since 2019 Optics Express, Optical Engineering

Session Co-Chair

2025 ICRA, the session "Visual Perception and Learning"

2025 CVPR, 1st International Workshop on Spatial Intelligence for Cultural Heritage, Oral Session 1 (with Rocco Pietrini).

Invited Talks

2026 **Dream to Assembly: Rebuilding the Broken by Imagining the Whole**.
Voxel51 AI, ML and Computer Vision Meetup

2025 **Learning Generalizable Models for Fragmented Cultural Heritage**.
CVPR, 1st International Workshop on Spatial Intelligence for Cultural Heritage

2025 **AI-Powered Understanding of Ancient Languages**.
NYU ArchaeoHack 2025

2025 **Could Foundation Models Uncover the Hidden Story of Ancient Materials?**
Paleoanthropology Society 2025 Annual Meeting; NYU Abu Dhabi Institute.

Mentorship

I have mentored Ph.D., M.S., and undergraduate researchers at NYU. Collaborative projects include GARF, CRAG, FusionSense, Wanderland, and the LUWA Dataset.